
VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION OF VISION–LANGUAGE MODELS

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ABSTRACT

Vision–Language Models (VLMs) show promise as zero-shot goal-conditioned value functions, but their frozen pre-trained representations limit both generalization and temporal reasoning. This paper makes three contributions that together address these two limitations. First, we introduce VITA, a zero-shot value function learning method that improves both capabilities through test-time adaptation: at inference, a lightweight adaptation module is updated by a single gradient step on a meta-learned self-supervised loss, so that each test-time update directly improves value estimation, and by applying these updates sequentially over a trajectory, VITA encodes history into its parameters and thereby overcomes the temporal reasoning limitations of prior work. Second, to prevent shortcut learning, we propose a dissimilarity-based sampling strategy that selects semantically diverse trajectory segments during training. Third, we validate these contributions empirically: on real-world robotic manipulation tasks, VITA generalizes from a single training environment to diverse out-of-distribution tasks, environments, and embodiments, outperforming the state-of-the-art zero-shot method built on autoregressive VLMs; and VITA’s zero-shot value estimates serve as reward shaping for offline reinforcement learning, yielding multi-task policies on the Meta-World benchmark that exceed those trained with the simulation’s fuzzy-logic dense rewards.

1 INTRODUCTION

Vision-Language Models (VLMs) have demonstrated strong generalization across diverse tasks and domains by learning from large-scale, unstructured web data without human supervision (Radford et al., 2021; Alayrac et al., 2022). In contrast, despite significant advances in learning generalist policies for robotics (Brohan et al., 2023; Kim et al., 2024) and 3D virtual environments (Reed et al., 2022; SIMA Team et al., 2024), state-of-the-art methods have yet to achieve comparable success, largely because they still rely on expert demonstrations (Ho & Ermon, 2016). World models (Ha & Schmidhuber, 2018; Hafner et al., 2021; Matsuo et al., 2022) have emerged as a promising alternative, enabling agents to learn in simulated environments, with applications spanning robotics (Yang et al., 2023; Agarwal et al., 2025), autonomous driving (Hu et al., 2024), and video games (Bruce et al., 2024). Yet even as world models generate realistic visual trajectories in diverse, open-ended environments (Hughes et al., 2024; Silver & Sutton, 2025), one central question remains open and motivates this work: *how can agents effectively learn from videos at scale without relying on human supervision?*

One line of research answers this question by learning policies directly from expert visual trajectories, inferring actions from visual transitions (Edwards et al., 2019; Baker et al., 2022; Schmidt & Jiang, 2024; Zhang et al., 2025a); the resulting latent actions can be mapped to executable controls at deployment, though doing so reliably remains difficult in practice. A second line, which we build on, instead learns a universal *goal-conditioned value function* that serves as a zero-shot reward shaping and supervision signal for reinforcement learning (RL) and imitation learning, respectively (Chen et al., 2021; Ma et al., 2022). In this framework, goal-conditioned value estimation is cast as predicting how far an agent has progressed toward completing a task, given visual observations and a natural language task description (Lee et al., 2021a; Ma et al., 2023b; 2024; Dashora et al., 2025). It is this second framework that our contributions target.

To motivate our contributions, we first identify a shared weakness in existing VLM-based value functions. Prior work employs pre-trained contrastive VLMs (Radford et al., 2021) for zero-shot reward shaping, exploiting the similarity between task descriptions and visual observations in a shared multimodal representation space to ground progress in semantic context (Ma et al., 2023b; Baumli et al., 2023; Rocamonde et al., 2024). These methods, however, fail to capture the temporal context needed to disambiguate visually similar states that occur at different stages of a task (e.g., folding vs. unfolding a shirt). Autoregressive VLMs (Alayrac et al., 2022) do incorporate temporal context by conditioning on the entire visual trajectory within the prompt, but they inherit a bias toward monotonically increasing predictions from the chronologically ordered datasets used in pre-training (Ma et al., 2024). Crucially, both architectures depend on *frozen* pre-trained representations (Ma et al., 2023b; 2024; Baumli et al., 2023; Rocamonde et al., 2024; Du et al., 2023) for zero-shot prediction, which simultaneously limits their generalization and their temporal reasoning (Pătrăucean et al., 2023). Attempts to remove these limits have so far leaned on large-scale pre-training (Alakuijala et al., 2024; Ma et al., 2023b), domain-specific fine-tuning (Zhang et al., 2025b), or demo-based reward specification (Sontakke et al., 2023)—each of which reintroduces a dependence on scale or supervision. This gap is the starting point for our method.

To close this gap, we introduce VITA, a zero-shot value function learning method that enhances both generalization and temporal reasoning through test-time adaptation, and that outperforms the state-of-the-art zero-shot approach (Ma et al., 2024) on real-world robotic manipulation tasks. The central idea is that, unlike prior methods that rely on large-scale pre-training or expert demonstrations for generalization, VITA adapts *online* to both the semantic and temporal context of each trajectory. Concretely, at inference a lightweight adaptation module is updated in negligible time with a gradient step on a meta-learned self-supervised loss (Sun et al., 2024), such that each test-time update improves value function estimation (Finn et al., 2017); and by applying these updates sequentially over a test trajectory, the estimator encodes trajectory history directly into its parameters (Sun et al., 2024), which is precisely what addresses the temporal reasoning limitation identified above. To keep this adaptation from latching onto shortcuts (Geirhos et al., 2020), we further introduce a dissimilarity-based sampling strategy that encourages reliance on semantic cues, a choice we support with empirical evidence. We validate VITA across the three core capabilities expected of a value function estimator: generalization under distribution shifts, differentiation between expert and non-expert trajectories, and reward shaping for offline RL (Levine et al., 2020).

In summary, this paper makes the following contributions:

- **A test-time adaptation method for value estimation.** We propose VITA, which enhances the generalization and temporal reasoning of contrastive VLMs for zero-shot value function estimation, without requiring task-specific demonstrations or large-scale pre-training.
- **A dissimilarity-based sampling strategy.** To prevent the adaptation mechanism from exploiting temporal shortcuts, we introduce a sampling strategy that selects semantically diverse trajectory segments during training, and we show it improves the ability to distinguish expert from non-expert trajectories.
- **Generalization across distribution shifts.** We demonstrate that VITA generalizes from a single training environment to diverse out-of-distribution tasks, environments, and embodiments in robotic manipulation, outperforming the state-of-the-art zero-shot method (Ma et al., 2024).
- **Zero-shot reward shaping for offline RL.** We show that VITA’s zero-shot value estimates, used as reward shaping, yield offline RL policies on the Meta-World multi-task benchmark (MT10) that surpass those trained with the simulation’s fuzzy-logic dense rewards (Yu et al., 2020).

2 PRELIMINARIES

2.1 GOAL-CONDITIONED VALUE FUNCTIONS

For video trajectories, task progress estimation can be viewed as goal-conditioned value function estimation, where the reward reflects task completion; we adopt this view throughout. Concretely, we formulate learning a goal-conditioned value function as predicting the degree of task completion from

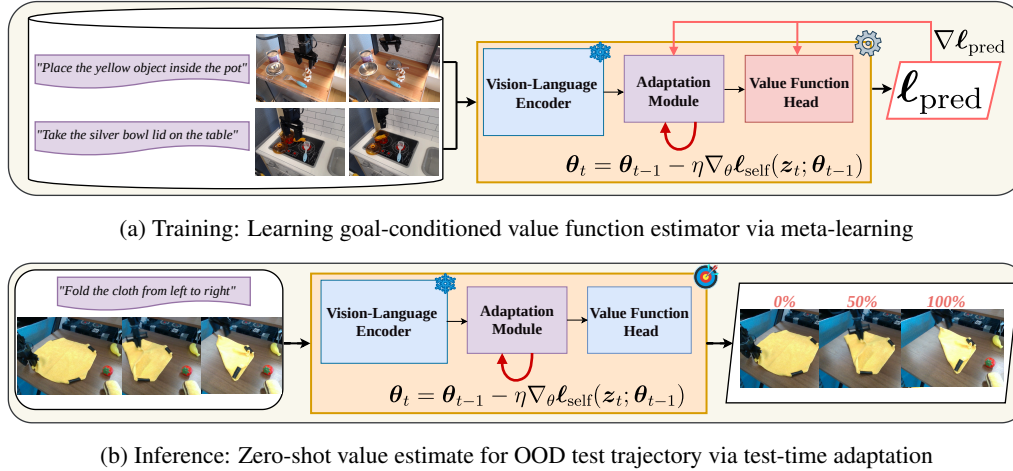


Figure 1: **Overview of VITA, illustrating our core contribution.** VITA learns a goal-conditioned value function via meta-learning (top) and achieves zero-shot generalization to out-of-distribution trajectories via test-time adaptation (bottom)—the two ingredients that jointly deliver the generalization and temporal reasoning gains reported in this paper.

vision–language representations. Formally, the value function $V : \mathcal{O} \times \mathcal{G} \rightarrow [0, 1]$ maps an observation $o_t \in \mathcal{O}$ and a goal specification $g \in \mathcal{G}$ to a scalar indicating predicted progress toward goal completion, with $V(o_t; g) = 0$ at the start of the task and $V(o_t; g) = 1$ at completion. Following common practice, task progress is aligned with temporal position in expert demonstrations, under the assumption that such trajectories exhibit monotonically increasing progress toward goal completion (Lee et al., 2021a; Ma et al., 2024; Dashora et al., 2025). Given an expert trajectory $\tau = (o_1, \dots, o_T) \sim \pi_E$, temporal progress is defined using normalized timestep indices: $V^{\pi_E}(o_t; g) = \frac{t}{T}$, where T is the trajectory length. Temporal progress thus supplies the supervision signal for learning a goal-conditioned value function V .

2.2 TEST-TIME TRAINING FOR SEQUENCE MODELING

Test-time training (TTT) (Sun et al., 2020) is the adaptation mechanism our method builds on. TTT treats inference as a self-supervised learning task (Goodfellow et al., 2016), updating model parameters on each test instance without access to labels. Originally proposed for static image classification (Sun et al., 2020), TTT has since been extended to sequence modeling (Sun et al., 2024; Wang et al., 2023), where a parametric adaptation module f_{adapt} is updated at each timestep by a gradient step on a self-supervised loss. Unlike standard sequence architectures that encode temporal history in activation states—for example, hidden states (Hochreiter & Schmidhuber, 1997; Cho et al., 2014) or cached key–value pairs (Vaswani et al., 2017)—TTT encodes history into the *parameters* of f_{adapt} via sequential updates at inference. These parameters therefore act as an implicit memory that preserves temporal order. Sun et al. (2024) additionally meta-learn the self-supervised task in the gradient-based meta-learning paradigm (Finn et al., 2017). In this setting (Sun et al., 2024), the self-supervised loss is parameterized by learnable linear projections and trained to minimize downstream prediction loss after a test-time update, rather than being minimized directly. We take this machinery in a new direction, applying TTT to goal-conditioned value function learning: temporal context is captured through sequential updates, and the self-supervised task is meta-learned specifically to improve value estimation rather than being fixed a priori.

3 VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION

3.1 MODEL ARCHITECTURE

We now describe the model that realizes our first contribution. Our goal-conditioned value function estimator comprises three modules: (1) a multimodal encoder that extracts joint visual-language

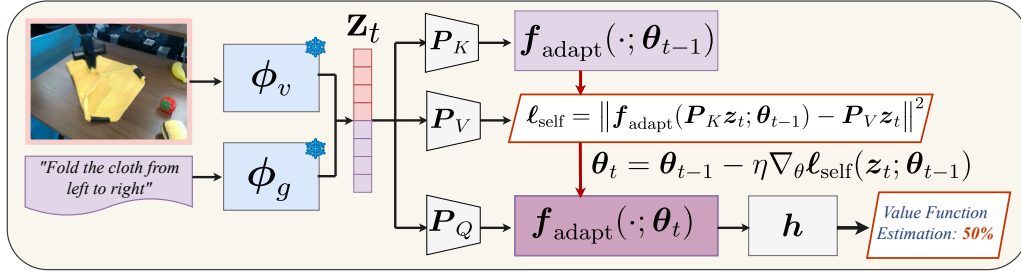


Figure 2: **Test-time adaptation, the mechanism behind VITA’s temporal reasoning.** At inference, at each timestep t , the adaptation module f_{adapt} is updated by a gradient step on a meta-learned self-supervised loss ℓ_{self} , so that trajectory history is progressively encoded into its parameters.

representations from visual trajectories and their task descriptions; (2) an adaptation module updated at test time using a self-supervised loss that is meta-learned to improve value estimation—the component responsible for VITA’s generalization and temporal reasoning; and (3) a regression head that predicts value estimates. We detail each in turn below.

3.1.1 MULTIMODAL INPUT REPRESENTATION

We use a frozen contrastive vision-language encoder, CLIP (Radford et al., 2021), to extract representations from visual observations and goal descriptions. Given a visual trajectory $\tau = (o_1, \dots, o_T)$ and a language task description g , we concatenate their representations at each timestep t to form a sequence of joint multimodal representations $(z_1, z_2, \dots, z_T)$, where $z_t = [\phi_v(o_t); \phi_g(g)] \in \mathbb{R}^{2d}$, and ϕ_v and ϕ_g denote the visual and language encoders, respectively. Because CLIP is pre-trained with a contrastive objective that aligns paired image-text inputs in a shared representation space (Radford et al., 2021), observations that are semantically closer to goal completion tend to lie closer to the goal description—the semantic signal our adaptation module exploits.

3.1.2 TEST-TIME ADAPTATION

At the heart of VITA is an adaptation module f_{adapt} that adapts multimodal representations to both semantic and temporal context, following the test-time training paradigm of 2.2. At inference, the parameters of f_{adapt} are updated online via a gradient step on a meta-learned self-supervised loss ℓ_{self} . This loss is a reconstruction objective built from learnable linear projections P_K and P_V , which are meta-learned during training so that adaptation improves value estimation. Concretely, given $P_K \in \mathbb{R}^{d' \times 2d}$ that generates a perturbed input view and $P_V \in \mathbb{R}^{d' \times 2d}$ for the target, the self-supervised loss is defined as:

$$\ell_{\text{self}}(z_t; \theta_{t-1}, P_K, P_V) = \|f_{\text{adapt}}(P_K z_t; \theta_{t-1}) - P_V z_t\|^2. \quad (1)$$

where θ_{t-1} denotes the parameters of f_{adapt} before test-time adaptation. At each timestep t , the parameters are updated by

$$\theta_t = \theta_{t-1} - \eta \nabla_{\theta} \ell_{\text{self}}(z_t; \theta_{t-1}), \quad (2)$$

with learning rate η . Through these sequential updates, f_{adapt} implicitly retains information from past visual observations and thereby captures temporal context—the key property that lets VITA reason over time. We substantiate this design choice in Section 4.6.2, where we show empirically that incremental, per-timestep parameter updates outperform updates performed at the trajectory level.

3.1.3 ZERO-SHOT VALUE FUNCTION ESTIMATOR

After test-time adaptation, a meta-learned linear projection $P_Q \in \mathbb{R}^{d' \times 2d}$ maps the input z_t into an adaptation space $\mathbb{R}^{d'}$. Then, this representation is passed through the adaptation module f_{adapt} , followed by a regression head h that outputs predicted values of the goal-conditioned value function:

$$V(z_t; g) = h(f_{\text{adapt}}(P_Q z_t; \theta_t)) \quad (3)$$

The regression head h is a two-layer multilayer perceptron (MLP), following value-function estimator architectures used in deep reinforcement learning (Espeholt et al., 2018). It is trained with normalized progress labels y_t from expert demonstrations (Section 2.1), under a supervised prediction loss ℓ_{pred} given by the mean squared error between predicted values $V(z_t; g)$ and targets y_t . Crucially for zero-shot use, at inference h remains frozen while only the adaptation module f_{adapt} is updated online via the self-supervised objective. This division of labor is what allows the estimator to generalize zero-shot, without any task-specific demonstrations.

3.2 TRAINING PROCEDURE

Training brings together the two methodological contributions of the paper. We train the value function estimator with gradient-based meta-learning, optimizing the self-supervised loss ℓ_{self} so that test-time adaptation improves the supervised prediction loss ℓ_{pred} (Sun et al., 2024). Training runs on diverse sub-trajectories selected by our dissimilarity-based sampling strategy, which mitigates shortcut learning and encourages reliance on semantic and temporal cues. We describe the meta-learning objective and the sampling strategy in turn.

3.2.1 META-LEARNING

During training, the adaptation module f_{adapt} is updated online at each timestep using the self-supervised loss ℓ_{self} . Following the gradient-based meta-learning paradigm (Finn et al., 2017), we differentiate through these adaptation updates with respect to the supervised prediction loss ℓ_{pred} , jointly optimizing the initialization θ_0 of f_{adapt} , the linear projections P_K, P_V, P_Q , and the regression head h . The total training loss combines ℓ_{pred} with ℓ_{self} , weighted by a scalar λ , and the learned θ_0 serves as the initialization for adaptation at inference. The upshot is a value function estimator that has *learned how to adapt* its internal representations at test time so as to improve value estimation—the mechanism underlying our zero-shot generalization results.

3.2.2 DISSIMILARITY-BASED SAMPLING

Our second contribution addresses a concrete failure mode of the training procedure above. Expert visual trajectories often contain highly redundant consecutive frames, as noted in prior work on video action recognition (Wang et al., 2018), and this redundancy can lead the value function estimator to exploit shortcut cues (Geirhos et al., 2020; Lee et al., 2021b) by overfitting to frequently occurring late-stage visual patterns. To counter this, we propose a dissimilarity-based sampling strategy that constructs each mini-batch from the most visually diverse sub-trajectories within a trajectory. This increases intra-batch variance and acts as a form of importance sampling that favors semantic diversity.

Formally, given a sequence of multimodal representations $\{z_1, \dots, z_T\}$, we extract fixed-length sub-trajectories using a sliding window of size w_{tr} and stride s , yielding a candidate set $\mathcal{W}$. A diverse subset of size k maximizes pairwise dissimilarity:

$$\mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{\{w^i, w^j\} \in \binom{\mathcal{U}}{2}} \|w^i - w^j\|_2^2, \quad (4)$$

where each w^i is a sub-trajectory of length w_{tr} . Solving Eq. 4 is intractable, requiring enumeration of all $\binom{|\mathcal{W}|}{k}$ subsets. To approximate this objective efficiently, we adopt a scoring-based heuristic, assigning each window $w \in \mathcal{W}$ a score equal to its total dissimilarity with all others:

$$s(w) = \sum_{v \in \mathcal{W}} \|w - v\|_2^2, \quad \mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{w \in \mathcal{U}} s(w). \quad (5)$$

Thus, by computing a diversity score s for each window and selecting the k highest-scoring windows, we obtain a heuristically diverse subset in polynomial time, with overhead that is negligible relative to model training (see the complexity analysis in Appendix G). We validate this contribution directly: the ablation in Section 4.6.1 shows that dissimilarity-based sampling improves the ability to distinguish expert from non-expert visual trajectories relative to full-trajectory sampling.

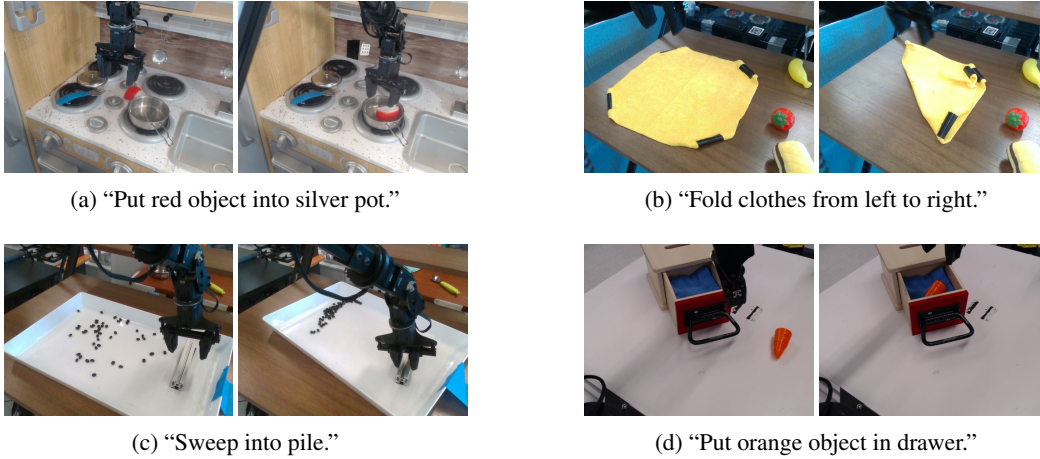


Figure 3: Examples of visual trajectories paired with task descriptions, spanning the distribution shifts against which we test VITA’s generalization contribution. (a) In-distribution. (b, c) Environment shift. (d) Embodiment and environment shift.

4 EXPERIMENTS

Our experiments are organized to test each claimed contribution in turn, evaluating VITA across the core capabilities of a goal-conditioned value function: (i) **generalization** in value estimation for real-world robotic manipulation under distribution shifts in task, environment, and embodiment; (ii) **robustness**, i.e., differentiation between expert and non-expert visual trajectories in real-world robotic manipulation tasks, which also validates our sampling contribution; (iii) **utility**, i.e., effective zero-shot reward shaping for offline RL in Meta-World MT10, a simulated multi-task robot learning benchmark. Ablation studies then isolate the two design choices behind these results—dissimilarity-based sampling and per-timestep test-time adaptation.

4.1 TRAINING SETUP

A key premise of our generalization claim is that VITA learns from a *single, narrow* training distribution. We train on expert visual trajectories paired with natural language task descriptions from BridgeData V2 (Walke et al., 2023), using a curated subset of 2,986 expert demonstrations that covers pick-and-place manipulation only—deliberately excluding folding, sweeping, and stacking tasks so that these later appear strictly out-of-distribution. All demonstrations use a single robot embodiment (WidowX 250) across 4 configurations of the ToyKitchen environment, and an additional 287 demonstrations are held out as the in-distribution test set `tk_pnp` (see Appendix A for examples). We use OpenCLIP ViT-B/32 as the frozen backbone for encoding video frames and task descriptions. The training objective combines ℓ_{pred} with ℓ_{self} , weighted by $\lambda_{\text{self}} = 0.5$, and training uses our dissimilarity-based sampling with window size $w_{\text{tr}} = 8$, number of sub-trajectories $k = 8$, and stride $s = 1$. At test time we take a single gradient step ($t_{\text{ep}} = 1$) using λ_{self} with learning rate $\eta = 0.1$. Full details and hyperparameter sweeps appear in Appendix D.

4.2 BASELINES

To situate our contributions, we compare VITA against value functions spanning both contrastive and autoregressive VLMs, including the current state of the art. VLM-CL computes zero-shot value estimates from the cosine similarity between CLIP representations of each frame and the task description (Baumli et al., 2023). VLM-RM regularizes CLIP embeddings by projecting features along the direction from a generic reference prompt to the task prompt (Rocamonde et al., 2024). CLIP-FT trains a supervised regression head on frozen CLIP multimodal representations, and CLIP-GRU trains a Gated Recurrent Unit (GRU) (Cho et al., 2014) on the same representations, serving as an explicit-memory temporal baseline. The most important comparison is GVL (Ma et al., 2024), the state-of-the-art zero-shot value function based on autoregressive VLMs, which we run

in two settings: GVL-0S (zero-shot; the VLM is prompted with only the test trajectory and task description) and GVL-1S (one-shot in-context; a full shuffled training-distribution trajectory with its progress labels is supplied as an example). Following GVL, we use Gemini 1.5 Pro (gemini-1.5 Pro-latest) (Gemini Team et al., 2024) with its proposed prompt template; we also tested Qwen-VL 2.5 (Yang et al., 2024), which failed to overcome temporal bias, and GPT-4o (Hurst et al., 2024), which refused to produce value estimates. Additional implementation details are provided in Appendix D.

4.3 EVALUATING GENERALIZATION UNDER DISTRIBUTION SHIFTS

Table 1: **Generalization contribution.** VOC scores for value function estimation under distribution shifts; higher is better, and best per row is in **bold** (second best underlined). VITA attains the best or second-best score on every dataset, and the best on 6 of 10, supporting our claim of generalization from a single training environment. **ID** = In-Distribution, **ES** = Environment Shift, **EM** = Embodiment Shift, **ES & EM** = Both Shifts.

Shift	Dataset	VLM-CL	VLM-RM	CLIP-FT	GVL-0S	GVL-1S	CLIP-GRU	VITA
ID	tk_pnp	0.038	0.029	0.251	0.269	0.252	<u>0.773</u>	0.782
	lm_pnp	0.017	0.033	0.149	0.305	0.272	<u>0.676</u>	0.725
ES	td_fold	0.031	0.072	0.152	0.326	0.318	<u>0.674</u>	0.709
	ft_fold	0.108	0.099	0.162	0.331	0.387	0.693	0.658
	rd_fold	0.095	0.055	0.126	0.372	0.406	0.726	<u>0.606</u>
	ms_sweep	-0.129	-0.226	0.148	0.158	0.150	<u>0.434</u>	0.490
EM	dt_tk_pnp	0.042	-0.041	0.149	0.258	0.211	0.856	<u>0.820</u>
	dt_tk_stack	0.035	0.046	0.099	0.254	0.277	<u>0.667</u>	0.708
ES & EM	dt_ft_stack	0.026	0.028	0.049	0.212	0.249	<u>0.674</u>	0.698
	dt_rd_pnp	0.023	0.041	0.211	0.329	0.316	0.747	<u>0.695</u>

This experiment directly tests our generalization contribution: can a value function trained on a single narrow distribution transfer to novel environments, tasks, and embodiments? We use subsets of BridgeData V2 curated to introduce variation along each axis. Environment shifts alter scene layout or background—for example, `lm_pnp` is pick-and-place in front of a laundry machine, while `td_fold`, `ft_fold`, and `rd_fold` involve cloth folding on different surfaces, and `ms_sweep` is a long-horizon sweeping task in a confined tray. Embodiment shifts use the DeepThought robot, which differs from the training embodiment (WidowX 250) in both morphology and camera perspective: `dt_tk_pnp` (pick-and-place) and `dt_tk_stack` (stacking) keep the in-distribution environment with the new embodiment, while `dt_ft_stack` (stacking) and `dt_rd_pnp` (drawer pick-and-place) combine both shifts. Each dataset contains 200 expert trajectories with task descriptions, except `ms_sweep` (100, due to its size), and Appendix B describes all evaluation datasets in full. We measure performance with the *Value Order Correlation* (VOC) metric (Ma et al., 2024), which quantifies how well predicted progress aligns with the chronological order of frames in a trajectory: letting $p_1, \dots, p_T$ be the predicted progress values for a trajectory of T frames and $r_t = t$ the temporal index, VOC is the Spearman rank correlation ρ_s between the two.

Table 1 supports our generalization contribution. Both VITA and CLIP-GRU consistently outperform GVL-0S and GVL-1S, underscoring the limitations of pre-trained autoregressive VLMs and the importance of preserving temporal order for progress estimation. Among the two temporal methods, VITA is the stronger, winning on 6 of 10 evaluation datasets. Both also outperform the CLIP-based baselines that lack temporal modeling (CLIP-FT, VLM-CL, VLM-RM), confirming that encoding temporal history matters for progress estimation. In detail, CLIP-FT matches GVL in-distribution but fails to generalize out-of-distribution, while VLM-CL and VLM-RM perform poorly throughout, likely for want of temporal modeling. For GVL, in-context examples help in some cases, but provide no benefit or even hurt in others: both GVL variants do well on folding tasks (`td_fold`, `ft_fold`, `rd_fold`) yet fall short on stacking (`dt_tk_stack`, `dt_ft_stack`) and pick-and-place (`dt_tk_pnp`, `lm_pnp`), suggesting a bias toward folding-like manipulations in the underlying autoregressive VLM. VITA, by contrast, is consistent across all task types and shifts—the central evidence that it generalizes more robustly than in-context learning.

Two settings sharpen this conclusion. On the long-horizon task `ms_sweep`, every method except CLIP-FT degrades, yet VITA still attains the highest VOC and surpasses CLIP-GRU—direct evidence that encoding temporal history via test-time adaptation beats recurrent hidden states for long-horizon manipulation. Under embodiment shift, VITA again generalizes strongly: on `dt_tk_pnp`, which changes only the robot embodiment, both VITA and CLIP-GRU even exceed their in-distribution performance, showing that learned temporal context transfers across embodiments, whereas GVL-1S performs poorly on both `dt_tk_pnp` and `dt_tk_stack`, indicating that few-shot in-context learning fails to generalize under embodiment shift in our setup.

4.4 DIFFERENTIATING EXPERT FROM NON-EXPERT TRAJECTORIES

Beyond generalization, our second evaluation targets robustness, and in doing so also stress-tests the sampling contribution: can VITA tell expert from suboptimal trajectories, assigning the latter lower progress scores? We compare model predictions on expert versus scripted (non-expert) visual trajectories drawn from the same in-distribution setting of BridgeData V2. The scripted trajectories are produced in the ToyKitchen environment by a heavily randomized controller (Walke et al., 2023), under the same embodiment and environment as the expert demonstrations, and cover pick-and-place tasks whose object categories may overlap with training: general objects (`sc_pnp_obj`), rigid items (`sc_pnp_robj`), soft toys (`sc_pnp_stoy`), utensils (`sc_pnp_uten`), and cloth (`sc_pnp_cloth`). Although these trajectories match the training distribution in task, embodiment, and environment, their behavior departs from the smooth, monotonic progress of expert demonstrations (see Appendix C for examples). Because the randomized controller is not designed to produce quantifiable degrees of suboptimality, we ask only that an effective estimator assign lower VOC to suboptimal than to expert trajectories in the same setting. We capture this with *BinVOC*, a binary metric $1 [\text{VOC}_{\text{exp}} > \text{VOC}_{\text{subopt}}]$ that is 1 when expert trajectories score higher than suboptimal ones and 0 otherwise.

We report BINVOC averaged over all scripted datasets, with expert performance (VOC_{exp}) measured on the in-distribution test set `tk_pnp`, whose pick-and-place tasks resemble those in the scripted evaluation. As summarized in Table 2(a), VITA, GVL-0S, and GVL-1S achieve perfect discrimination, always scoring expert trajectories above suboptimal ones. The most informative comparison is against CLIP-GRU: VITA outperforms it, indicating that recurrent hidden-state memory is more prone to overfitting temporal shortcuts than the implicit memory formed by sequential test-time updates—and thereby corroborating the value of our sampling contribution. The CLIP-based baselines (VLM-CL, VLM-RM) underperform, again for lack of temporal modeling, and CLIP-FT, while more reliable, fails on one task, revealing sensitivity to object distribution shift.

4.5 ZERO-SHOT REWARD SHAPING FOR OFFLINE RL

Our third contribution is that VITA’s zero-shot value estimates are useful as reward shaping for offline RL. We test this on the Meta-World MT10 benchmark (Yu et al., 2020), ten diverse robotic manipulation tasks including pick-and-place, door opening, and pushing. The setup deliberately stresses transfer: VITA is trained on real-world robotic data and applied zero-shot in this simulated setting. For each MT10 task we generate 20 expert visual demonstrations with Meta-World’s expert policies, use the goal-conditioned value estimator to define a dense reward for each visual observation, and train policies with Implicit Q-Learning (IQL) (Kostrikov et al., 2021) in the offline RL regime. We repeat demonstration generation and policy training over 10 random seeds, evaluate each multi-task policy on 20 rollouts per task, and record the average return per (task, seed) pair. Following Agarwal et al. (2021), we report the interquartile mean (IQM) over all pairs with 95% stratified bootstrap confidence intervals; further details are in Appendix E.

The result, in Table 2(b), confirms the contribution: VITA achieves the highest IQM return (0.815), outperforming all CLIP-based baselines. A direct comparison with GVL is infeasible at this scale, as it relies on Gemini-1.5—a proprietary autoregressive VLM with prohibitive per-step inference cost in RL—and open-source alternatives did not yield reliable progress estimates in our preliminary tests. Most tellingly, VITA also surpasses Meta-World’s own fuzzy-logic dense reward (META-WL, 0.779), showing that a value estimator trained purely on real-world data can generalize to simulated reward shaping and even improve on a hand-designed simulator reward.

Method	BinVOC
VLM-CL	0.40
VLM-RM	0.00
CLIP-FT	0.80
GVL-0S	1.00
GVL-1S	1.00
CLIP-GRU	0.80
VITA	1.00

(a) Expert vs. Non-Expert (BinVOC).

Method	IQM	95% CI
VLM-CL	0.760	[0.722, 0.791]
VLM-RM	0.746	[0.718, 0.771]
CLIP-FT	0.785	[0.759, 0.809]
CLIP-GRU	0.777	[0.734, 0.814]
META-WL	0.779	[0.750, 0.804]
VITA	0.815	[0.785, 0.838]

(b) Offline RL on MT10 (IQM).

Table 2: Evidence for the robustness and utility contributions. (a) Success in distinguishing expert from scripted robot demonstrations, measured by average BinVOC across 5 in-distribution scripted datasets: VITA attains a perfect 1.00, matching the GVL variants and exceeding CLIP-GRU. (b) Offline RL performance on the Meta-World MT10 benchmark, measured by interquartile mean (IQM) with 95% stratified bootstrap CIs: VITA obtains the highest IQM, surpassing every baseline including the simulator’s fuzzy-logic reward (META-WL).

4.6 ABLATION STUDIES

4.6.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

This ablation isolates our second contribution, quantifying how much the dissimilarity-based sampling strategy contributes to VITA’s ability to distinguish expert from non-expert demonstrations. We compare three sub-trajectory sampling strategies. Full-trajectory sampling (Sun et al., 2024) computes the adaptation loss over all overlapping sub-trajectories with stride $s = 1$ and, in our setup, overfits to global temporal shortcuts. Random sampling adds diversity by selecting sub-trajectories uniformly, but does not target semantic diversity. Our dissimilarity-based sampling instead selects sub-trajectories that maximize pairwise dissimilarity, explicitly promoting semantic diversity. The takeaway is clear: our strategy outperforms both full-trajectory and random sampling at differentiating expert from non-expert visual trajectories, confirming that the sampling contribution is responsible for a measurable part of VITA’s robustness (details in Appendix F.1).

4.6.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

This ablation isolates the mechanism behind our first contribution, examining how the choice of test-time adaptation strategy affects value function estimation. VITA performs step-by-step adaptation without reset (implicit memory), incrementally updating parameters from θ_{t-1} so that the estimator encodes trajectory history directly into its parameters, capturing both temporal order and accumulated context. We compare VITA against three alternatives: (i) *trajectory-level adaptation (TTT-TR)*, which applies a single gradient update computed over the full trajectory. This produces a batched update that averages the loss across all frames, thereby discarding temporal order; (ii) *memoryless adaptation (TTT-RS)*, which resets the adaptation module to θ_0 at every step. By adapting only to the current visual observation, this approach prevents any temporal information from being carried forward; (iii) *explicit memory (TTT-EX)*, which also resets the adaptation module at every step but updates using a local window of recent frames matching the window size used during meta-training ($w_{tr} = 8$). The result is unambiguous: across all datasets, VITA consistently outperforms the trajectory-level, memoryless, and explicit-memory variants, establishing that incrementally accumulating temporal context—precisely the design choice at the core of our contribution—is what drives value estimation quality. We provide the full analysis in Appendix F.2.

5 RELATED WORK

5.1 VLMS AS VALUE FUNCTION ESTIMATORS

We position our contributions against two families of VLM-based value functions. Contrastive VLMs have served as zero-shot reward models (Baumli et al., 2023; Rocamonde et al., 2024) and goal-conditioned value functions (Ma et al., 2023b; 2024), using frame-level similarity scores but

without temporal modeling. Others fine-tune CLIP on domain-specific video datasets (Fan et al., 2022; Jiang et al., 2024) to produce dense rewards, at the cost of supervision that limits zero-shot applicability; RoboCLIP (Sontakke et al., 2023) enables one-shot reward specification without fine-tuning but yields sparse rewards and still needs demonstrations; and large-scale multimodal pre-training approaches (Alakuijala et al., 2024; Li et al., 2024; Ma et al., 2023b) learn language-conditioned value functions by relying on scale for generalization. ReWiND (Zhang et al., 2025b) trains a cross-modal sequential transformer on augmented large-scale data with video rewinding plus a few in-domain demonstrations. The distinguishing feature of our contribution is that VITA instead adapts a pre-trained contrastive encoder *at inference*, capturing temporal and semantic context through test-time adaptation without large-scale multimodal pre-training or domain-specific demonstrations—and, being agnostic to the encoder, it could enhance the generalization and temporal reasoning of any pre-trained vision–language representation that encodes a universal value function. The second family, autoregressive VLMs (Alayrac et al., 2022), uses in-context learning over trajectories for success detection (Du et al., 2023) and goal-conditioned value functions (Ma et al., 2024). Here GVL (Ma et al., 2024) mitigates the monotonic bias of ordered pre-training data by shuffling frames at inference, but shuffling discards the temporal order that is essential for temporal reasoning and for separating visually similar yet temporally distinct states. Our contribution takes the opposite route: it *preserves* chronological order while capturing temporal context through test-time adaptation, thereby avoiding these shortcuts.

5.2 TEST-TIME ADAPTATION

Test-time adaptation offers parameter-efficient ways to adapt VLMs without full fine-tuning, and our contribution builds on one such mechanism. Prompt-based methods optimize parameters at test time from task context, though their mechanisms remain opaque (Zhou et al., 2022; Khattak et al., 2023; Ma et al., 2023a). Lim et al. (2025) argued that CLIP representations can support test-time adaptation, motivated by the observation that CLIP encodes human-interpretable concepts identified via mechanistic interpretability (Bricken et al., 2023), and CLIP-based similarity has been used as a reward whose parameters are optimized at test time via reinforcement learning (Zhao et al., 2024). We instead adopt test-time training (Sun et al., 2020), in which an adaptation module is updated at each timestep via a self-supervised loss. Unlike meta-reinforcement learning (Finn et al., 2017; Bauer et al., 2023), which requires task-specific adaptation episodes during training, test-time training adapts directly online at inference without task labels—the property that makes it the natural foundation for a zero-shot value function.

6 DISCUSSION, LIMITATIONS, AND FUTURE WORK

6.1 DISCUSSION

Taken together, our results substantiate each contribution claimed in the introduction. First, test-time adaptation enables value functions for robotic manipulation to generalize across distribution shifts in task, environment, and embodiment. Second, VITA distinguishes expert from non-expert visual trajectories and performs zero-shot reward shaping, opening downstream applications in RL and imitation learning. The mechanism uniting these outcomes is that, by updating sequentially over a trajectory, VITA encodes history into its parameters, capturing temporal and semantic context more effectively than CLIP-based baselines or in-context learning with autoregressive VLMs. That VITA beats CLIP-GRU on both expert–non-expert discrimination and offline RL is direct evidence that implicit memory via sequential test-time updates generalizes better than explicitly storing temporal history in recurrent hidden states. Pre-trained autoregressive VLMs (GVL), for all their generalization, are not trained for progress estimation, which demands temporal reasoning over trajectories. Finally, these gains come at negligible inference-time cost: only a lightweight adaptation module is updated per timestep, so real-time applicability is preserved.

6.2 LIMITATIONS AND FUTURE WORK

We close by delineating the boundaries of our contributions. While test-time adaptation improves zero-shot progress estimation across unseen tasks and environments in our experiments, it can still struggle under high execution variability or very long horizons. And although the adaptation

overhead is negligible, updating a value function estimator at every timestep may be unsafe in some deployments, which could limit real-time use. These open issues point to several directions. We plan to study alternative test-time adaptation schemes for vision–language models, especially for training agents within world models, and to apply VITA to real-time closed-loop control and more complex RL environments; because VITA is agnostic to the underlying encoder, it could also extend generalization and temporal reasoning to any domain-specific pre-trained multimodal representation. Finally, our evidence that dissimilarity-based sampling mitigates shortcut learning is empirical, and a theoretical account of how diversity-based sampling shapes shortcut learning is a natural next step—one that lies beyond, but complements, our core contribution: a value function estimator that meta-learns a test-time adaptation mechanism to improve zero-shot performance and temporal reasoning.

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A TRAINING DATASET

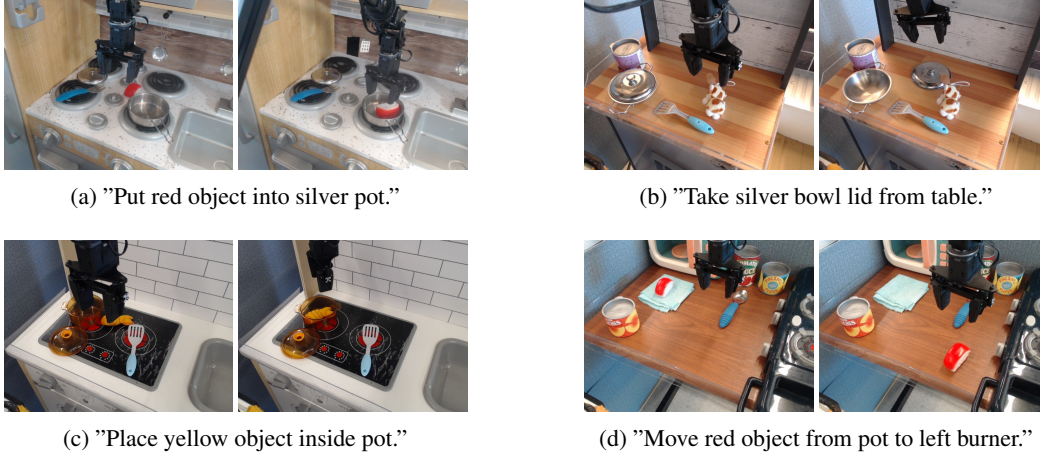


Figure 4: Start and end frames of expert demonstrations used for training, each with its natural language task description. All demonstrations are collected across four distinct ToyKitchen environments, illustrating the deliberately narrow single-environment training distribution from which VITA must generalize.

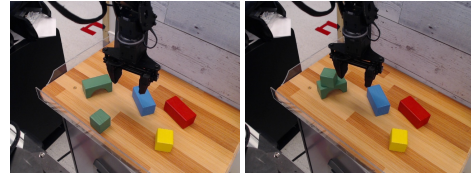
B EVALUATION DATASETS

Table 3: Evaluation datasets used to test the generalization contribution, listed by task type, environment, and embodiment.

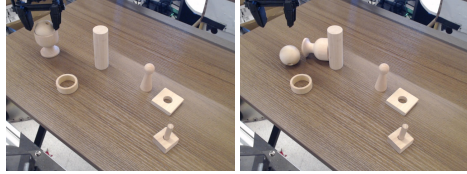
Dataset	Task Type	Environment	Embodiment
tk_pnp	pick-and-place	toy kitchen	WidowX 250
lm_pnp	pick-and-place	laundry machine	WidowX 250
td_fold	fold cloth	tabletop (dark wood)	WidowX 250
ft_fold	fold cloth	folding table	WidowX 250
rd_fold	fold cloth	robot desk	WidowX 250
ms_sweep	sweep	folding table (tray)	WidowX 250
dt_tk_pnp	pick-and-place	toy kitchen	DeepThought
dt_tk_stack	stack blocks	toy kitchen	DeepThought
dt_ft_stack	stack blocks	folding table	DeepThought
dt_rd_pnp	pick-and-place	robot desk (drawer)	DeepThought



(a) dt_tk_ms: "Move yellow object from burner to center."



(b) dt_tk_stack: "Place green cube on top of arch."

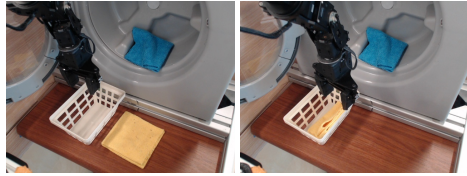


(c) dt_ft_stack: "Move egg to table."



(d) dt_rd_pnp: "Put orange object in drawer."

Figure 5: Start and end frames of evaluation trajectories under embodiment shift, each with its natural language task description. The top row keeps the training environment (ToyKitchen) but uses a different robot (DeepThought), while the bottom row also introduces new environments—the harder combined-shift setting for our generalization claim.



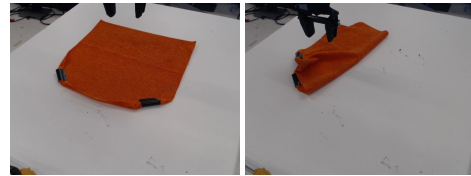
(a) lm_pnp: "Place blue cloth inside washer."



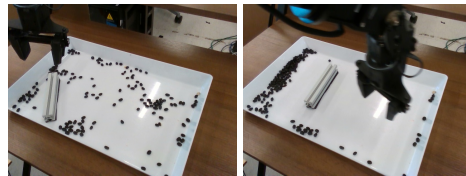
(b) td_fold: "Fold cloth bottom right to top left."



(c) ft_fcnp: "Fold clothes from left to right."



(d) rd_fold: "Fold cloth bottom right to top left."



(e) ms_ft_sweep: "Sweep into pile."

Figure 6: Start and end frames of evaluation trajectories under environment shift, each with its natural language task description. All tasks use the same robot embodiment across visually and structurally different environments, isolating the environment axis of our generalization evaluation.

C SCRIPTED DATASET



(a) `sc_pnp_obj`: "Pick up an object and place it."



(b) `sc_pnp_robj`: "Pick up a rigid object and place it."



(c) `sc_pnp_stoy`: "Pick up a soft object and place it."



(d) `sc_pnp_uten`: "Pick up a kitchen utensil and place it."



(e) `sc_pnp_cloth`: "Move the cloth."

Figure 7: Start and end frames of scripted evaluation trajectories collected in the ToyKitchen environment with the WidowX 250 robot, used to test the expert-vs.-non-expert discrimination behind our robustness contribution. All tasks are pick-and-place with varying object categories, and we use generic task descriptions because per-trajectory annotations are not publicly available.

D MODEL ARCHITECTURE

D.1 MODEL AND TRAINING DETAILS

For completeness, we record the configuration behind the results reported above. We use the OpenCLIP ViT-B/32 encoder pre-trained on the OpenAI dataset as a frozen backbone, encoding each visual observation and task description with the CLIP vision and text encoders, respectively. We adopt a joint (concatenation-based) representation over an element-wise product, a choice made on the basis of improved validation performance. The adaptation module f_{adapt} is a two-layer residual MLP with GELU activation and projection dimension $d' = 64$. Training uses the AdamW optimizer with learning rate 1×10^{-4} , weight decay 1×10^{-4} , and a cosine schedule with 10% warmup, a batch size of 32, and all trajectories padded to a maximum length of 120 frames (the longest sequence in the training set). The self-supervised weighting coefficient λ_{self} is set to 0.5, again selected on validation performance, and we train for 5 epochs, since validation VOC plateaus early and longer training yields no gain. For dissimilarity-based sampling we use stride $s = 1$, sub-trajectory length $w_{\text{tr}} = 8$, and $k = 8$ selected windows unless stated otherwise. All experiments ran on NVIDIA RTX 6000 Ada Generation GPUs.

D.2 TEST-TIME TRAINING HYPERPARAMETERS

At inference we adapt only the temporal adaptation module f_{adapt} , keeping the same projection dimension ($d' = 64$) as in training, and perform a single gradient step ($t_{\text{ep}} = 1$) with learning rate $\eta = 0.1$. This configuration was chosen from a sweep over learning rates $\{5.0, 1.0, 0.1, 0.05, 0.01, 0.001\}$

Table 4: Sensitivity of VITA to the test-time learning rate η and number of adaptation steps t_{ep} , measured by validation VOC; this sweep motivates the single-step, $\eta = 0.1$ configuration used throughout.

η	$t_{\text{ep}} = 1$	$t_{\text{ep}} = 5$	$t_{\text{ep}} = 10$
0.001	0.225	0.345	0.453
0.01	0.454	0.744	0.781
0.05	0.746	0.670	0.515
0.1	0.783	0.512	0.369
1.0	0.299	0.183	0.181
5.0	0.169	0.182	0.157

and adaptation steps $\{1, 5, 10\}$ on the validation set (Table 4). Two settings emerge as comparably strong—a single update with a larger learning rate ($\eta = 0.1$, $t_{\text{ep}} = 1$) and multiple updates with a smaller one ($\eta = 0.01$, $t_{\text{ep}} = 10$)—and we adopt the former, as it reaches slightly better validation performance with fewer adaptation steps.

D.3 BASELINES

To ensure the comparisons supporting our contributions are fair, we detail each baseline here. VITA uses $t_{\text{ep}} = 1$, projection dimension $d' = 64$, and learning rate $\eta = 0.1$, tuned as described above. CLIP-FT mirrors VITA’s architecture but drops the adaptation module and self-supervised loss: a frozen CLIP encoder feeds a linear projection and a two-layer MLP that predicts task progress, trained by standard supervised regression without meta-learning or test-time adaptation, and—to give it every advantage in expressivity—it uses an $8\times$ larger projection matrix and $10\times$ more training steps than our method (Lin et al., 2022). For VLM-RM we use a baseline prompt that describes the environment. CLIP-GRU likewise follows VITA’s architecture (frozen CLIP encoder and regression head h) but replaces f_{adapt} with a Gated Recurrent Unit (GRU) that encodes temporal history *explicitly* through recurrent hidden states; it is a single layer of hidden size 64 whose final hidden state is passed to h , and it is the sharpest point of comparison for our implicit-memory design. For both GVL-0S and GVL-1S we use the latest Gemini 1.5 Pro, `gemini-1.5-pro-latest` (Gemini Team et al., 2024), rather than the earlier `gemini-1.5-pro` of the original GVL implementation; we also evaluated the open-source `Qwen-VL 2.5` (Yang et al., 2024), which failed to overcome temporal bias despite frame shuffling, while GPT-4o (Hurst et al., 2024) consistently declined to produce scalar progress estimates in our setup. All evaluations followed the prompt template from the original GVL work (Ma et al., 2024).

E OFFLINE RL SETUP

E.1 TRAINING CONFIGURATION

So that our reward-shaping contribution can be reproduced, we fix a single Implicit Q-Learning (IQL) configuration across all tasks. Expectile regression is set to $\tau = 0.7$, with advantage-weighting temperature 3.0 and clipping threshold 100.0; the actor and critic use learning rates 1×10^{-4} and 3×10^{-4} with batch size 256. Policies train for 100,000 gradient steps and are evaluated every 10,000 steps on 20 rollouts per task (horizon 150). The implementation follows `d3rlpy` with configuration `expectile=0.7`, `weight_temp=3.0`, `max_weight=100.0`, `actor_lr=1e-4`, `critic_lr=3e-4`, `batch.size=256`.

E.2 EVALUATION PROTOCOL

To keep the utility comparison statistically sound, we follow Agarwal et al. (Agarwal et al., 2021) and report the interquartile mean (IQM) of returns across all task–seed pairs. The IQM averages the middle 50% of scores, reducing sensitivity to outliers relative to the mean or median, and we estimate 95% confidence intervals via stratified bootstrap with 10,000 resamples, stratified by task. This protocol is applied identically to every method in the Meta-World MT10 experiments.

F ABLATION STUDIES

F.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

Table 5: **Dissimilarity-based sampling contribution.** Disc@VOC across scripted datasets under different training-time sampling strategies for test-time adaptation; a checkmark marks successful discrimination between expert and suboptimal trajectories. Our dissimilarity-based sampling with $(w_{tr}, k) = (8, 8)$ attains perfect discrimination, whereas the full-trajectory baseline is worst. Sampling methods **FT** (full-trajectory), **Rand** (random), and **Diss** (dissimilarity-based) are parameterized by (w_{tr}, k) .

Dataset	FT (8,-)	Rand (8,8)	Diss (8,4)	Diss (4,8)	Diss (8,8)
sc_pnp_obj		✓	✓		✓
sc_pnp_robj	✓	✓	✓	✓	✓
sc_pnp_stoy			✓		✓
sc_pnp_uten	✓	✓	✓	✓	✓
sc_pnp_cloth		✓	✓	✓	✓
Avg. Disc@VOC	0.20	0.8	1.0	0.60	1.0

This appendix expands the validation of our sampling contribution, comparing how different training-time sub-trajectory sampling strategies affect the model’s ability to discriminate expert from suboptimal trajectories. In Table 5 we report Disc@VOC on 5 scripted datasets, indicating whether a model assigns higher average VOC to expert trajectories (`tk_pnp`) than to each evaluation dataset. We contrast three strategies: (1) full-trajectory sampling, which adapts over all overlapping sub-trajectories of length w_{tr} with stride s , following prior work (Sun et al., 2024); (2) random sampling, which draws k sub-trajectories of length w_{tr} uniformly per video; and (3) our dissimilarity-based sampling, which builds a candidate set of sub-trajectories of length w_{tr} with stride s and selects the k that maximize pairwise dissimilarity in representation space.

The results confirm the contribution and clarify its sensitivities. Dissimilarity-based sampling with $w_{tr} = 8$ and $k = 8$ yields perfect discrimination; fewer windows ($k = 4$) at $w_{tr} = 8$ match it, while shortening sub-trajectories to $w_{tr} = 4$ degrades performance. The full-trajectory baseline is worst, confirming that adapting over all consecutive windows overfits to global temporal cues, and random sampling, though better than full-trajectory, remains less reliable than dissimilarity-based selection. Together these findings show that both sub-trajectory diversity and semantic locality are needed for discriminative test-time adaptation—the two properties our sampling strategy is designed to provide.

F.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

To pinpoint why VITA’s adaptation mechanism works, we compare four adaptation strategies: (1) TTT-TR, which performs a single gradient update computed over the full trajectory (offline). TTT-TR performs a batched update that averages the loss across all frames, thereby discarding temporal order; (2) TTT-RS, which resets the adaptation module at every step and updates using only the current visual observation (memoryless). TTT-RS prevents any temporal information from being carried forward; (3) TTT-EX, which resets the adaptation module at every step and updates using a local window of recent frames matching the window size used during meta-training ($w_{tr} = 8$). TTT-EX averages gradients over this window, discarding the temporal ordering within it; (4) VITA (OURS), which updates the adaptation module incrementally at every timestep (implicit memory). VITA forms an implicit memory through the sequential accumulation of temporal history in the adapter parameters. We performed the same test-time hyperparameter sweeps for all variants, as described in Appendix D. We found that VITA, TTT-RS, and TTT-TR achieve optimal performance with a learning rate of 0.1 and a single adaptation step, whereas TTT-EX performs best with a learning rate of 1.0 and one adaptation step.

Table 6 tells a sharp story. TTT-TR, TTT-RS, and TTT-EX perform comparably, meaning that batched updates—whether over a short explicit window (TTT-EX) or the full trajectory (TTT-TR)—are no better than the memoryless baseline (TTT-RS). VITA consistently outperforms all three, which is the strongest evidence that the sequential, per-timestep updates at the heart of our contribution are

Table 6: **Temporal-memory ablation supporting our core contribution.** Validation VOC for progress estimation under distribution shifts across adaptation strategies; VITA’s per-timestep implicit memory wins on every dataset by a wide margin. ID = In-Distribution, ES = Environment Shift, EM = Embodiment Shift, ES+EM = Environment and Embodiment Shift.

Shift	Dataset	TTT-TR	TTT-RS	TTT-EX	VITA
ID	tk_pnp	0.1917	0.1918	0.1957	0.7822
ES	lm_pnp	0.1843	0.1854	0.1826	0.7246
	td_fold	0.1402	0.1393	0.1322	0.7085
	ft_fold	0.1302	0.1311	0.1683	0.6583
	rd_fold	0.1161	0.1172	0.1154	0.6056
	ms_sweep	-0.0225	-0.0191	0.0102	0.4898
EM	dt_tk_pnp	0.2123	0.2118	0.2069	0.8203
	dt_tk_stack	0.0833	0.0830	0.0747	0.7081
ES+EM	dt_ft_stack	0.0558	0.0537	0.0439	0.6979
	dt_rd_pnp	0.1665	0.1660	0.1559	0.6951

what make the mechanism effective for real-world robotic manipulation. The reason is structural: in VITA’s implicit memory, updating over a local window would count the most recent observations twice (explicitly in the window and implicitly in the accumulated parameters), while updating less frequently would drop intermediate frames and lose information. The implicit-memory test-time training paradigm therefore *requires* sequential, per-timestep updates without reset.

G COMPLEXITY ANALYSIS OF DISSIMILARITY-BASED SAMPLING

Here we show that the sampling contribution introduces negligible cost. Let $\mathcal{W}$ denote the candidate set of size $N_c = |\mathcal{W}|$. To approximate diverse subset selection without the exponential cost of enumerating all $\binom{N_c}{b}$ subsets of Eq. 4, we use a scoring-based heuristic that assigns each candidate window a diversity score equal to the sum of its pairwise distances to all others (the row sum of the distance matrix). With $N_c = T - w_{tr} + 1$ candidate windows, each a flattened feature vector of dimension $m = w_{tr} \cdot d$ (where d is the dimension of each multimodal representation z_t), we compute the full pairwise distance matrix $D \in \mathbb{R}^{N_c \times N_c}$ on the GPU via batched `cdist`, at a cost of $\mathcal{O}(N_c^2 \cdot w_{tr} \cdot d)$ operations.

In our experiments the training trajectories are short (mean length 69, max length 120). With batch size 32, feature dimension $d = 1024$ (the CLIP-based multimodal representation), and window length $w_{tr} = 8$, the worst-case cost is:

$$32 \cdot N_c^2 \cdot w_{tr} \cdot d = 32 \cdot 114^2 \cdot 8 \cdot 1024 \approx 340 \text{ MFLOPs.}$$

This overhead is negligible next to a single forward–backward pass through our model, so the sampling contribution promotes diverse-window selection at essentially no added cost.